

CS-4112: Introduction to Machine Learning

Statistics

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Expectation

- ▶ Expectation is a weighted average of a function.
- ▶ Weights are given by $p(x)$.

$$\mathbb{E}[f] = \sum_x p(x)f(x) \quad \longleftarrow \text{For discrete } x$$

$$\mathbb{E}[f] = \int_x p(x)f(x)dx \quad \longleftarrow \text{For continuous } x$$

- ▶ When data is finite, expectation $\approx$ ordinary average. Approximation becomes exact as $N \rightarrow \infty$ (*Law of large numbers*).

Expectation

- ▶ Expectation of a function of several variables

$$\mathbb{E}[f(x, y)] = \sum_{x, y} p(x, y) f(x, y)$$

- ▶ Expectation with respect to one variable

$$\mathbb{E}_x[f(x, y)] = \sum_x p(x) f(x, y) \quad (\text{function of } y)$$

- ▶ *Conditional expectation*

$$\mathbb{E}_{x|y}[f] = \sum_x p(x|y) f(x)$$

Variance

- ▶ *Variance* measures variability of a random variable around its mean.

$$\begin{aligned}\text{var}[f] &= \mathbb{E}[(f(x) - \mathbb{E}[f(x)])^2] \\ &= \mathbb{E}[f(x)^2] - \mathbb{E}[f(x)]^2\end{aligned}$$

- ▶ On average, how far does a random variable stay from its mean?

Covariance

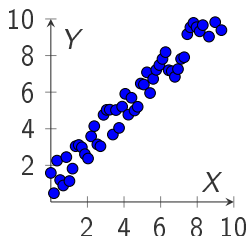
Univariate

- ▶ For 2 univariate random variables occurring in pairs (x, y) , **covariance** expresses how much x and y vary together.

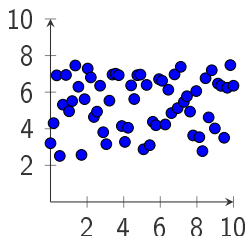
$$\begin{aligned}\text{cov}[x, y] &= \mathbb{E}_{x,y} [\{x - \mathbb{E}[x]\}\{y - \mathbb{E}[y]\}] \\ &= \mathbb{E}_{x,y} [xy] - \mathbb{E}[x] \mathbb{E}[y]\end{aligned}$$

- ▶ For independent random variables x and y , $\text{cov}[x, y] = 0$. Why?

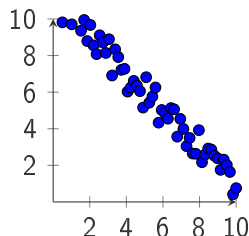
Positive Covariance



Zero Covariance



Negative Covariance



Covariance

Multivariate

- ▶ For multivariate random variables $\mathbf{x} \in \mathbb{R}^D$ and $\mathbf{y} \in \mathbb{R}^K$, $\text{cov}[\mathbf{x}, \mathbf{y}]$ is a $D \times K$ matrix.
- ▶ Expresses how each element of $\mathbf{x}$ varies with each element of $\mathbf{y}$.

$$\begin{aligned}\text{cov}[\mathbf{x}, \mathbf{y}] &= \mathbb{E}_{\mathbf{x}, \mathbf{y}} \left[\{\mathbf{x} - \mathbb{E}[\mathbf{x}]\} \{\mathbf{y} - \mathbb{E}[\mathbf{y}]\}^T \right] \\ &= \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\mathbf{x} \mathbf{y}^T] - \mathbb{E}[\mathbf{x}] \mathbb{E}[\mathbf{y}]^T \\ &= \begin{bmatrix} \text{cov}[x_1, y_1] & \text{cov}[x_1, y_2] & \cdots & \text{cov}[x_1, y_K] \\ \text{cov}[x_2, y_1] & \text{cov}[x_2, y_2] & \cdots & \text{cov}[x_2, y_K] \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}[x_D, y_1] & \text{cov}[x_D, y_2] & \cdots & \text{cov}[x_D, y_K] \end{bmatrix} \end{aligned} \quad (1)$$

Covariance

Multivariate

- ▶ Covariance of multivariate $\mathbf{x}$ with itself can be written as $\text{cov}[\mathbf{x}] \equiv \text{cov}[\mathbf{x}, \mathbf{x}]$.
- ▶ $\text{cov}[\mathbf{x}]$ expresses how each element of $\mathbf{x}$ varies with every other element.

$$\text{cov}[\mathbf{x}] = \begin{bmatrix} \text{var}[x_1] & \text{cov}[x_1, x_2] & \cdots & \text{cov}[x_1, x_D] \\ \text{cov}[x_2, x_1] & \text{var}[x_2] & \cdots & \text{cov}[x_2, x_D] \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}[x_D, x_1] & \text{cov}[x_D, x_2] & \cdots & \text{var}[x_D] \end{bmatrix} \quad (2)$$

Bayesian View of Probability

- ▶ So far we have considered probability as the *frequency of random, repeatable events*.
- ▶ What if the events are not repeatable?
 - ▶ Was the moon once a planet?
 - ▶ Did the dinosaurs become extinct because of a meteor?
 - ▶ Will the ice on the North Pole melt by the year 2100?
- ▶ For non-repeatable, yet uncertain events, we have the **Bayesian view** of probability.

Bayesian View of Probability

$$p(\mathbf{w}|\mathcal{D}) = \frac{p(\mathcal{D}|\mathbf{w})p(\mathbf{w})}{p(\mathcal{D})}$$

- ▶ Measures the uncertainty in model $\mathbf{w}$ after observing the data $\mathcal{D}$.
- ▶ This uncertainty is measured via conditional $p(\mathcal{D}|\mathbf{w})$ and prior $p(\mathbf{w})$.
- ▶ Treated as a function of $\mathbf{w}$, the conditional probability $p(\mathcal{D}|\mathbf{w})$ is also called the **likelihood function**.
- ▶ Expresses how likely the observed data is for any given model $\mathbf{w}$.